

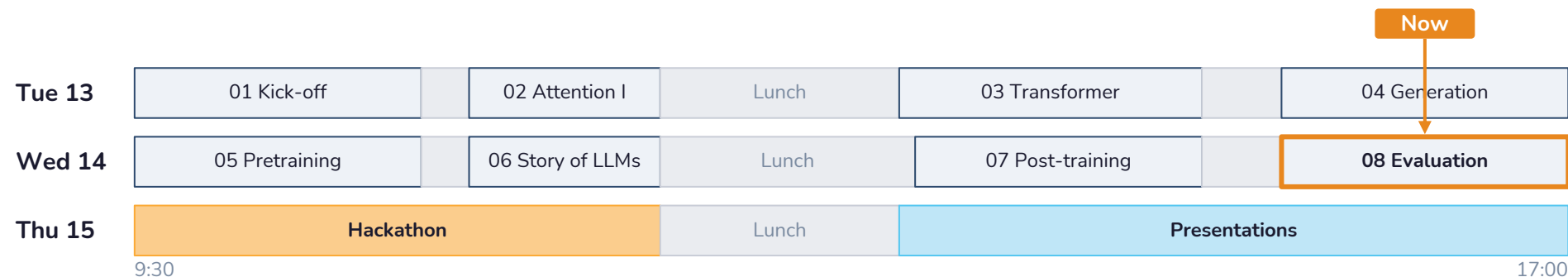
Advanced NLP

Session 08 • Evaluating LLMs

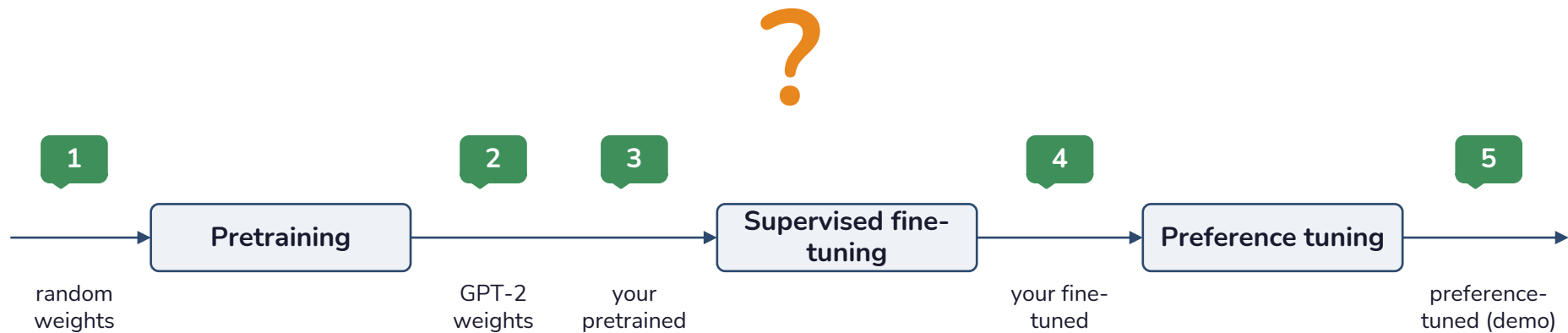
Prof. Dr. Richard Sieg • TH Köln IWS • WS 26 / 27
Wednesday, October 14, 2026

Which model is best, and what that means tomorrow

Time	
15:30	Two metrics, benchmarks, judges
15:59	Lab: notebook 08
16:19	Thursday's project
16:41	Your component
17:00	End of day 2



You have five models now. Which one is best?



A measurement only tells you what it measures, and only above the noise.

Agenda

01. Two metrics

02. Benchmarks

03. Judges

04. Thursday's project

01.

Two metrics

Perplexity says the pretrained model is best

Model	Perplexity on held-out TinyStories
Prétrainéð	5.12
SGT	5.61
DPÓ	5.89

- Lower is better. The pretrained model was trained on exactly this kind of text.
- Fine-tuning and preference tuning moved the model away from plain stories, towards following instructions.

The required-words rate says exactly the opposite

Móðél	Þérpléyitý	Rér úiréð!wórdś!úséð
Prétráinéð	5.12, bést	15.3%, wórst
SGT	5.61	44.7%
DPÓ	5.89, wórst	70.0%, bést

- Same three models, opposite rankings.
- Perplexity measures how well a model fits a kind of text. The word rate measures whether it does a task.

Neither number is “the” score. Each answers its own question.

02.

Benchmarks

MMLU: 57 knowledge tasks, multiple choice

- Massive Multitask Language Understanding
- Contains 57 diverse knowledge tasks
- Was the “go to” LLM benchmark until recently
- Benchmarks can’t keep up with LLM development

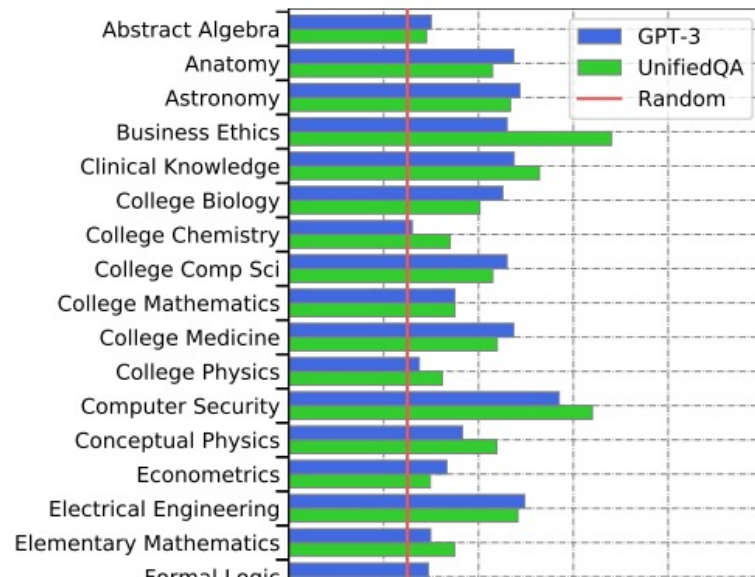
Why isn't there a planet where the asteroid belt is located?

- (A) A planet once formed here but it was broken apart by a catastrophic collision.
(B) There was not enough material in this part of the solar nebula to form a planet.
(C) There was too much rocky material to form a terrestrial planet but not enough gaseous material to form a jovian planet.
(D) **Resonance with Jupiter prevented material from collecting together to form a planet.**

Figure 16: An Astronomy example.

If each of the following meals provides the same number of calories, which meal requires the most land to produce the food?

- (A) Red beans and rice
(B) **Steak and a baked potato**
(C) Corn tortilla and refried beans
(D) Lentil soup and brown bread

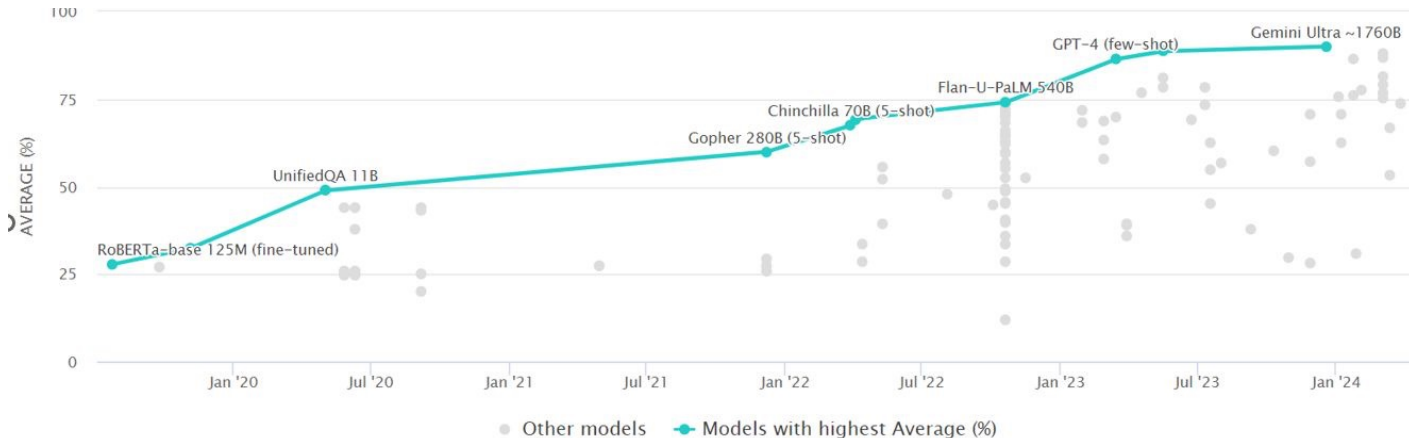
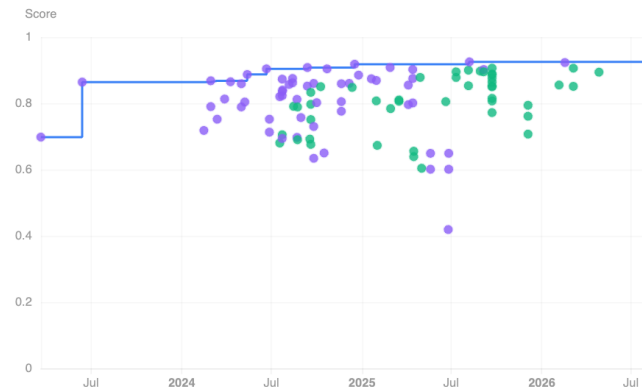


MMLU went from hard to nearly solved

[LLM Stats Leaderboard](#)

GPT3 was SOTA

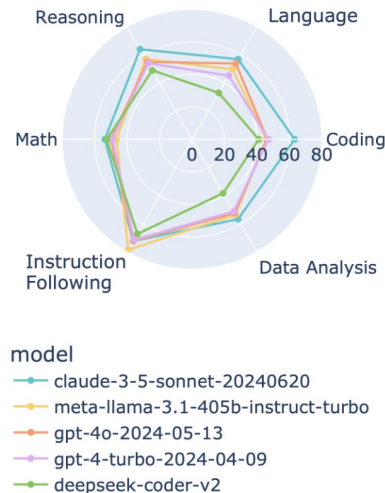
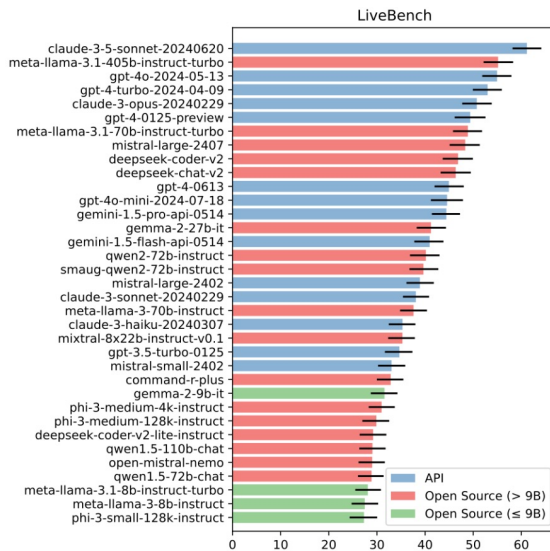
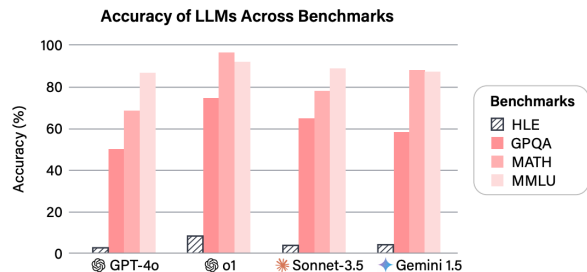
Model	Humanities	Social Science	STEM	Other	Average
Random Baseline	25.0	25.0	25.0	25.0	25.0
RoBERTa	27.9	28.8	27.0	27.7	27.9
ALBERT	27.2	25.7	27.7	27.9	27.1
GPT-2	32.8	33.3	30.2	33.1	32.4
UnifiedQA	45.6	56.6	40.2	54.6	48.9
GPT-3 Small (few-shot)	24.4	30.9	26.0	24.1	25.9
GPT-3 Medium (few-shot)	26.1	21.6	25.6	25.5	24.9
GPT-3 Large (few-shot)	27.1	25.6	24.3	26.5	26.0
GPT-3 X-Large (few-shot)	40.8	50.4	36.7	48.8	43.9



Measuring Massive Multitask Language Understanding [Hendrycks et al, 2020]

New benchmarks keep replacing the solved ones

- BIG-Bench (Google, 2022) contains 204 tasks
- Humanity's Last Exam (AI Safety, 2025)
- LiveBench (2024) gets updated monthly, to limit potential contamination into the training data of LLMs



Our model scores 25 percent on benchmarks: chance

- Benchmarks need scale. HellaSwag, ARC and the other standard suites: four answer options, written for models with billions of parameters and broad world knowledge.
- A 17.7 million parameter model that has only read children's stories scores at chance: 25 percent, whichever of the three checkpoints you pick.
- That is why nobody benchmarks a model this size, and why our metrics this week were loss and a task we could check.

Contamination inflates scores, saturation ends them

- **Contamination:** the benchmark's questions end up in the training data. The model recognizes answers instead of working them out, like a student who saw the exam.
- **Saturation:** every model scores near the top, and the benchmark no longer tells them apart.
- Assume any public dataset is or will be contaminated.
- Defenses: a canary string to search training data for, gated or encrypted test sets, benchmarks that change over time, like LiveBench.

Leaderboard numbers are hard to reproduce

- The same model scores differently with a different prompt template, a different number of examples in the prompt, or different examples.
- With sampling, the seed and the temperature matter. For reasoning models, whether thinking is switched on, and how many tokens it may use.
- Labs choose which checkpoint, which prompt and which run to report.

03.

Judges

An LLM can judge outputs when humans cannot keep up

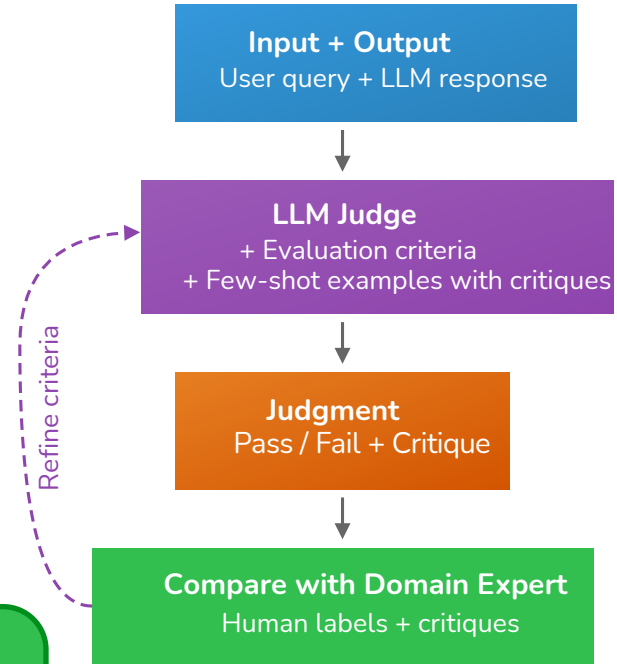
- Issue: Human annotation does not scale
- **LLM as a Judge**
 - LLM receives input, output, and evaluation criteria
 - LLM produces judgement and explanation (critique)

Common Pitfalls

- Position bias (prefers first answer)
- Length bias (prefers longer answer)
- Self-preference

Best Practices

- Binary pass/fail decision
- Include critique
- Use task-specific criteria



Judges have biases, each with a countermeasure

Bias	What happens	Countermeasure
Position	prefers the first, or the second, answer	swap the order at random
Verbosity	prefers longer answers	account for length
Self-preference	prefers outputs like its own	a jury of different judges
Inconsistency	changes its verdict when asked again	ask several times, take the majority
Format	judges badly outside its training format	follow the format it was trained on

Every metric this week could be gamed

- **Perplexity:** train on the test distribution.
- **The required-words rate:** repeat the words. Tom was slow, but he was slow.
- **A benchmark:** see its questions during training.
- **A judge:** write longer answers, or answer first.
- IFEval checks rules like word counts automatically, over hundreds of prompts: our required-words rule at scale, with the same weakness.

When a measure becomes a target, it stops being a good measure.



Tutorial

Notebook 08: measure the three models yourself

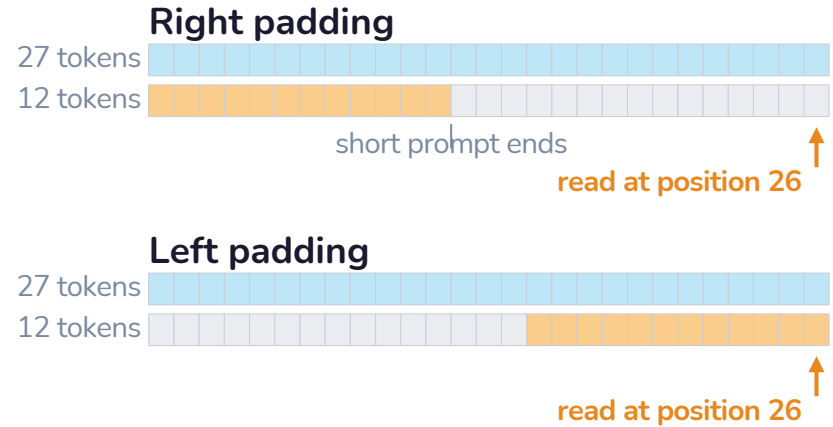
1. Implement `compute_perplexity`: build batches, average the loss, exponentiate.
2. Run the table for the three checkpoints. Do you get the same ranking?
3. Run the right-padding cell, then the left-padding one. Compare what each predicts for the short prompt.
4. Read three completions of the DPO model and three of the SFT model. Which would you rather read?

Abóút!20!mínútés#ín!páirís

- ☐ Your perplexity table matches the slide's ranking
- ☐ You can say why right padding reads the wrong position

The right-padding bug gives a plausible answer

- Two prompts of 27 and 12 tokens, padded on the right to 27.
- The model reads its next token at position 26 for both: 15 positions past the end of the short prompt.
- It predicts “Once” for both. A plausible start of a story, and still the answer to a question nobody asked.
- With left padding, each prompt ends at the last position: the fox story and a story about a boy called Jack, each continuing its own prompt.



A bug that produces plausible output is worse than one that crashes.

04.

Thursday's project

Thursday: does one change help, and can you tell?

GPT-2#ánđ!ýóúr!móðél	2026!ópén!móðél	Tħúrśđáy
LáýérNđrṃ	RMSNorm	01
Pré-LN	pre-LN, with post-LN variants back in OLMo 2	01
ĠELU!géeđ-ğrwarz	SwiGLU, often as a mixture of experts	02
Gúll!cáúsál!átténṭiún	sliding window in some layers	03
Ráw!r úérý-kéý!ścórés	QK-Norm	04
Léárnéđ!pósiṭiúnś	RoPE	05#06
Múłti-ħéáđ!átténṭiún	grouped-query attention	07
Ġénérátiún!wıthóút!á!cáħé	KV cache	08
Drópóút!0*1	no dropout	

Your folder: task, code, tests and harness

- `TASK.md`: what the component is, what it replaces, exactly what to implement, which runs and which measurements.
- `starter.py`: the classes and functions, named, with their arguments; the bodies are yours.
- `test_shapes.py`: run it first. It tells you the wiring is right, not that the component is.
- `test_sanity.py`: limit cases, for example “with the window as long as the context, sliding-window attention equals normal attention”.
- `ablation_harness.py` (one level up, shared by all components): runs `baseline.py` and your component under the fixed protocol and writes results in the format of `results_schema.json`.
- A reading list: the original paper, then blogs and code that start from the same GPT you built.

```
components/  
├── 03_sliding_window/  
│   ├── TASK.md  
│   ├── starter.py  
│   ├── test_shapes.py  
│   └── test_sanity.py  
├── ablation_harness.py  
├── baseline.py  
├── results_schema.json  
└── README.md
```

one of eight component folders, 01 to 08

Change one component, and nothing else

- **The baseline:** your GPT-2, the small configuration, unchanged, 2,000 steps, seeds 0, 1 and 2.
- **Your component:** the same 2,000 steps, the same seeds, the same data, learning rate, batch and context.
- **One change** means the component and everything it needs: RoPE also removes the learned positions. It is still one change.
- **Parameter count:** some components change it. Report it, do not compensate for it, discuss it.

Validation loss is the right metric here

A metric for training experiments should, according to the team behind FineWeb:

- ✓ score above chance
- ✓ improve as training goes on
- ✓ vary little between seeds
- ✓ rank models consistently

Validation loss on our fixed validation set does all four. Benchmarks, at our size, do none.

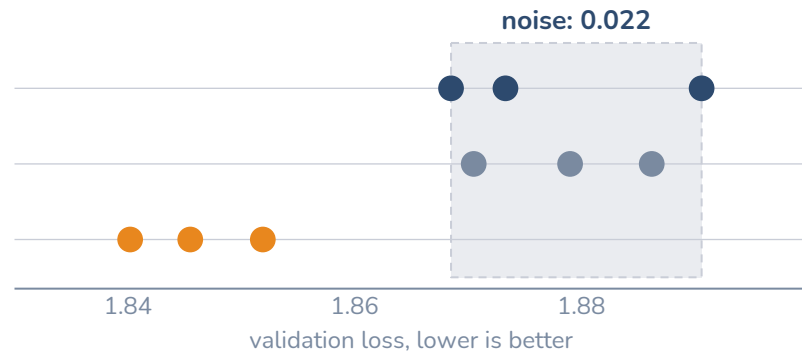
Smaller than 0.022 is not an effect

Seed	Baseline validation loss
0	1.8733
1	1.8906
2	1.8685
Spread	0.022

Baseline

Component A

Component B



A: inside the noise, inconclusive. B: outside it, an effect. Points for A and B are illustrative.

- Training is deterministic for a given seed: your three baseline runs should reproduce these numbers exactly.
- If your component moves the mean by less than 0.022, the honest answer is: inconclusive at this scale. That scores as well as a clear win.

An effect inside the noise is no effect.

Each component has a second metric
for what it was built to improve

Cómpónéńt	Śéćónď!métríc
ŘMŚNđrm#ńórm!plácémént	step time; training stability
ŚwíĜLŮ	loss at matched parameters
Ślđínĝ!wínđów	loss by position, and window size
RĶ-Nđrm	largest attention logit during training
Śínúsóidál!pósiťińś	loss by position
ŘóPE	loss by position
Ĝróúpéd-rúérý!áťťéńťiń	KV cache size
ĶV!ćáćhé	tokens per second, on GPU and CPU

Thursday is graded on the experiment, not its result

- **Presentation, Thursday afternoon, 60 percent:** 12 minutes per pair, then 6 minutes of questions, to one of you at a time, both of you asked.
- **Report and repository, October 21, 40 percent:** four pages, with a section on what did not work, and the appendix on AI use.
- Whether your component helps is not graded. Whether you can tell is.

Thursday, hour by hour

Time

9:30

Start your three baseline runs with the harness in your folder, on your second GPU, then a short briefing

9:45

Implement your component, run its shape and sanity tests

from 10:30

Component runs, at the same three seeds; a checkpoint with each pair

12:15

Lunch

13:30

Presentations, 18 minutes per pair

9:30



Hackathon

Lunch

Presentations

9:30

17:00

Thu 15

A measurement only tells you what it measures, and only above the noise.

Your component

Páir	Cómpónéint
1	
2	
3	
4	
5	
6	
7	
8	

Tonight: read your task, and one chapter

- Your component's `TASK.md`, and the first paper on its reading list.
- The ablation chapter of the Smol Training Playbook: how a team that trains real models runs exactly this kind of experiment.
- Run your shape test once. It should fail with “not implemented”, not with an import error.

Sources

Literature

- Fourier et al., The LLM Evaluation Guidebook, Hugging Face: contamination and saturation, reproducing published scores, judge biases and their mitigations, choosing evaluations for training ablations.
- Hendrycks et al. 2020, Measuring Massive Multitask Language Understanding, arxiv.org/abs/2009.03300.
- Zheng et al. 2023, Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena, arxiv.org/abs/2306.05685. Position bias.

Literature, continued

- Zhou et al. 2023, Instruction-Following Evaluation for Large Language Models, arxiv.org/abs/2311.07911. IFEval.
- Eldan, Li 2023, TinyStories, arxiv.org/abs/2305.07759. Grading by GPT-4.
- The Smol Training Playbook, Hugging Face. The ablation chapter.